OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL GCSE

BIOLOGY

9201/2

Paper 2

Mark Scheme

June 2019

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196x92012/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	body mass		1	AO2 3.5.3 3.6.1
01.2	one mark per correct response more than one line from a genetic term will cancel mark		3	AO1 3.5.3 c,i 3.6.1

Genetic term

- Chromosome
- DNA
- Variation

Description

- structure in the nucleus that contains genes
- a type of reproduction involving sex cells
- the differences in characteristics between organisms of the same kind
- having two different alleles for a characteristic
- the chemical genes are made from

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	bb		1	AO3 3.5.3
01.4	Bb bb	ignore colour description	1 1	AO2 3.5.3
01.5 View with 01.4	3:1	ecf from answer to 01.4	1	AO3 3.5.3 6.3(6)

01.6	one extra chromosome **or** three (copies) of chromosome 18 **or** 47 chromosomes (in total)		1	AO2 3.5.3d
01.7	Down's (syndrome)	allow Trisomy 21 allow other correctly named condition	1	AO1 3.5.4b
01.8	any **two** from: • more accurate • reduced chance of getting a wrong (positive) result • able to check / look at baby's chromosomes / DNA (for other abnormalities)	allow greater chance of getting a correct result	2	AO3 3.5.4b
01.9	any **two** from: • can be done earlier (on in pregnancy) • no risk of miscarriage **or** no harm to the mother / embryo • Test 2 accuracy is almost as high	allow it is safer allow <u>only</u> 2% less accurate allow it is easier to obtain (blood) samples ignore costs	2	AO3 3.5.4b
Total			**14**	

Question	Answers	Mark	AO/ Spec. Ref
02.1	**Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.	**4-6**	AO1 3.3.3e
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.	**1-3**	AO1 3.3.3e
	No relevant content	**0**	AO1 3.3.3e
	Indicative content Feeding • carbon is transferred from the plants to the animals during feeding • carbon (from the plants) becomes part of the organic compounds **or** carbohydrates **or** proteins **or** fats **or** named carbon compound in animals Photosynthesis • plants photosynthesise and use / absorb carbon dioxide (from the atmosphere) • carbon (from the carbon dioxide) is used to make organic compounds **or** carbohydrates **or** proteins **or** fats **or** named carbon compound • (which are) required for the growth of the plant Respiration • plants and animals respire using carbohydrates **or** proteins **or** fats **or** named carbon compound • carbon is released as carbon dioxide (into the atmosphere) Decay • dead plants and animals are decomposed by microorganisms • microorganisms respire using carbon compounds • carbon is released as carbon dioxide (into the atmosphere)		

02.2	to allow oxygen to enter		1	AO2 3.3.3b

02.3	any **one** from: • highest % for nitrate • highest % for phosphate • highest % (overall) of ions / minerals	allow concentration / amount ignore potassium use of values must be comparative	1	AO3 3.3.3c

02.4	type of manure		1	AO4 6.1

02.5	any **one** from: • (to act) as a control **or** for a comparison • to show the effect of the manure • so the results are valid	allow description do **not** accept it was a control variable	1	AO4 6.1

02.6		if no other mark awarded allow manure helps plants grow better / taller for one mark		AO3 3.3.3c
	rabbit manure gave greatest growth / height (in plants)		1	
	cow manure gave the least growth		1	
	cow and horse manure gave very similar results **or** growth / height of plants with horse manure were only slightly more than with cow manure	use of values must be comparative	1	

02.7	plants / wheat are removed (from the area / cycle) each year		1	AO2 AO3 3.3.3a
	(so) fewer mineral (ions) go back into the soil (from decomposition or addition of manure)	allow named example of mineral allow minerals are depleted / reduced (in soil)	1	
	(so) plants the next year have fewer (mineral ions) to make proteins	allow described function of named mineral in relation to reduced growth	1	
	to grow (and increase in mass)	allow to increase in height	1	

Total			**17**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	plasma	allow phonetic do **not** accept plasma proteins	1	AO1 3.2.3l
03.2	Any **two** from: • carbon dioxide • urea • glucose • amino acids • fatty acids • glycerol	allow other correctly named substance eg antibodies, hormones, mineral (salts), vitamins, ions, salts, lactic acid ignore oxygen ignore nutrients / food do **not** accept water, starch, glycogen, glucogen	2	AO1 3.2.3m
03.3		‘more’ must be included at least once for full marks		AO2 3.2.3n
	more room / space for haemoglobin (molecules)	ignore surface area	1	
	to transport / carry more oxygen (from lungs to organs)		1	

03.4	(platelets activate an) enzyme controlled reaction (which causes conversion of) fibrinogen to fibrin (which) produces a network / mesh of (protein) fibres which trap blood cells (forming a clot)	allow thrombin is produced / released	1 1 1	AO1 3.2.3q

03.5	(248.3 ÷ 5 =) 49.66 **and** (25.4 ÷ 5 =) 5.08 (49.66 – 5.08 =) 44.58 (%) **or** 44.6 (%)	an answer of 44.58% or 44.6% scores 2 marks ignore 45 (%)	1 1	AO2 3.2.3t 6.3(3,4,7)

03.6	any **three** from: • Chile has the least % of both A and B • India is the only country with more B than A • UK has the biggest difference between A and B • The mean % of A for the five countries is 28.78 **or** 28.8 compared with the mean % of B for the five countries is 16.48 **or** 16.5 • UK (and Japan) has the highest percentage of A **or** India has the highest percentage of B	allow overall (blood group) A has a higher total / mean than (blood group) B allow use of other correct comparative examples	3	AO2 AO3 3.2.3t

<table>
<tr>
<td>03.7</td>
<td>
<table>
<tr><th rowspan="2">Donor</th><th colspan="4">Recipient</th></tr>
<tr><th>A</th><th>B</th><th>AB</th><th>O</th></tr>
<tr><td>A</td><td>✓</td><td>X</td><td>✓</td><td>X</td></tr>
<tr><td>B</td><td>X</td><td>✓</td><td>✓</td><td>X</td></tr>
<tr><td>AB</td><td>X</td><td>X</td><td>✓</td><td>X</td></tr>
<tr><td>O</td><td></td><td></td><td></td><td></td></tr>
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</td>
<td>1 mark for each correct horizontal row

or

if no horizontal rows correct allow 1 mark for three vertical rows correct</td>
<td>3</td>
<td>AO2
3.2.3t</td>
</tr>
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03.8	donor red blood cells do not have either antigen A or B (on their surface)	allow the donor red blood cells have no antigen (on their surface)	1	AO2 3.2.3t	E
	(so) no reaction / binding between antigen (of donor) and (complementary) antibody in recipient / plasma	allow recipients antibodies are unable to attack antigens	1		
	no agglutination / clumping of red blood cells **or** no destruction of red blood cells	ignore prevents rejection ignore reference to blood clotting	1		

Total			**19**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	any **two** from: • length / 10cm of tubing • time /12 hrs left in solution • volume / 10 cm^3 of (sugar) solution put into tubing • (left at same) temperature/20°C	allow amount allow concentration of <u>unknown</u> (sugar) solution do **not** accept concentration unqualified	2	AO4 3.1.5 6.1
04.2	to remove any solution (from outside of tubing)	allow water for solution	1	AO4 3.1.5 6.1
04.3	$\frac{(-)1.85}{10.26} \times 100$ (–)18.03(%) (–)18(%)	an answer of (–)18(%) scores 3 marks allow correct rounding of an incorrectly calculated value	1 1 1	AO2 3.1.5 6.3(3)
04.4	X-axis – suitable scale and correctly labelled including units points plotted correctly suitable line of best fit should be a smooth curve	$> \frac{1}{2}$ axis spread of data allow axis label at bottom of graph allow tolerance of +/– ½ a small square allow 1 mark for 4 correct plots ignore plot calculated from 04.3 not point to point or feathery	1 2 1	AO2 3.1.5 6.3(9)

04.5	correct from candidate's graph	allow tolerance of +/– ½ a small square	1	AO3 3.1.5 6.3(11)	
04.6	repeat at smaller intervals between 0.6 (mol/dm^3) and 0.8 (mol/dm^3)	 allow a range which includes the value given in question 04.5	1 1	AO4 3.1.5 6.1	
04.7	any **one** from: • use a different balance • ensure balance was set at zero (before recording mass) • add / subtract the (consistent) error from the results	 allow calibrate the balance / reset the balance	1	AO4 3.1.5 6.1	E
04.8	it would give a reading / value (consistently) greater / less (than the true value) (therefore) an incorrect % change in mass or an incorrect sugar concentration (would be recorded)		1 1	AO4 3.1.5 6.1	E
04.9	decay is caused by microorganisms sugar solution is more concentrated than (cell contents of) microorganism (therefore) water moves out of microorganism by osmosis (so) microorganisms are unable to survive / reproduce (due to lack of water in their cells)	allow bacteria / fungi / decomposers allow sugar solution is more concentrated than (cell contents of) fruit allow (cell content of) fruit has a higher water potential allow (therefore) water moves out of fruit by osmosis	1 1 1 1	AO1 AO2 3.1.5d 3.3.3b	E

Total			**20**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	(each generation) used the claw more and more	allow 'use it or lose it' in relation to the claw	1	AO2 3.6.2a
	(so) the claw becomes larger / stronger		1	
	(and) then the larger / stronger claw is inherited (by the offspring)	ignore gene passed onto offspring	1	
05.2	any **six** from: • different flowering times • ability to survive in contaminated soil • isolate/ separate (the two populations) • genetic variation (in each population) **or** different / new alleles **or** mutation(s) (produced in each population) • natural selection occurs **or** better adapted survive • alleles passed on (for flowering time or survival in contaminated soil) • (eventually) resulting in successful interbreeding no longer possible **or** production of fertile offspring no longer possible (between the two populations)	allow geographical isolation (for isolate/ separate) allow gene / mutation (for alleles passed on)	6	AO1 AO2 3.6.2c
Total			**9**	

Question	Answers	Extra information	Mark	AO / Spec.
06.1	uses visual communication	allow sound communication	1	AO2 AO3 3.4.6a,e
	by making more prominent	allow description allow idea of throat vibrations / movements / flashes	1	
	which attracts a mate **or** as a warning sign for predators		1	

Question	Answers	Extra information	Mark	AO / Spec.
06.2	male attracted to the chemical (from the female) on the paper	allow scent / smell	1	AO2 AO3 AO4 3.4.6e
	airtight box prevents chemical from female moth reaching the male	allow chemicals / smell cannot get out of box	1	
	male does not use visual communication **or** the visual signal or sight of female was insufficient	allow male moth was not attracted by the female's appearance	1	

Question	Answers	Extra information	Mark	AO / Spec.
06.3	whales have difficulty communicating with others **or** whales lose sense of direction **or** whales are unable to navigate (and become stranded on beaches)	allow idea of confusion	1	AO3 3.4.6e
	sound (pollution) interferes with echoes		1	

	or loud sounds (could) damage their hearing			
	whales communicate using the same (frequency) range as the oil drilling / wind farms / large ships **or** oil drilling / wind farms / large ships produce sound in the (frequency) range from less than 0.6 (arbitrary unit) to just over 3.6 (arbitrary unit) which whales can detect / hear		1	
	some sound from submarines is in the same (frequency) range that whales use **or** (frequency) range from submarines of 2.0 (arbitrary unit) to 2.6 (arbitrary unit and/or 3.0 (arbitrary unit) to 4.0 (arbitrary unit) can be detected / heard by whales		1	
	there is correlation but no causation / proof so it is not definite		1	

Total			**11**